Worksheet 3: Limits and Continuous Functions

MATH 2263

09/15/2026

1 Determining Continuity

1.1 Graphically

For the following problems let $f(x)$ and $g(x)$ be functions whose graphs are given below. The graph of $f(x)$ is the solid graph, the graph of $g(x)$ is the dashed graph.

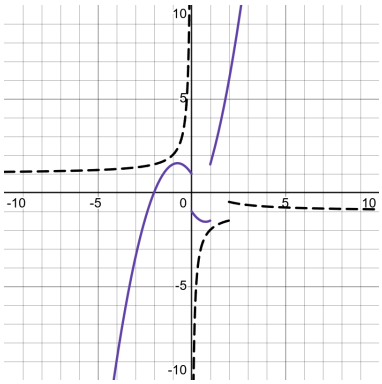


Figure 1: Function Graphs

Determine if the following functions are continuous as the indicated points:

1. Is $f(x)$ continuous at -2?
2. Is $f(x)$ continuous at -1?
3. Is $f(x)$ continuous at 0?
4. Is $g(x)$ continuous at 0?
5. Is $g(x)^2 + f(x)$ continuous at -2?
6. Is $f(x) + g(x)$ continuous at 2?
7. Is $f(x+1)^2 g(x-1)$ continuous at 0?

1.2 Algebraically

1. Is $f(x) = x^3 - 2x + 1$ continuous at $x = 0$? Why or why not?

2. Is $g(x) = \dfrac{x^2 - 1}{x + 1}$ continuous at $x = -1$? Why or why not?

3. Is $h(x) = \dfrac{\sin(x)}{x}$ a continuous function? Why or why not?

4. Is the function $F(x) = \begin{cases} -x, & x < 0 \\ 0, & x = 0 \\ x, & x > 0 \end{cases}$ a continuous function? Why or why not?

5. Is the function $G(x) = \begin{cases} x^3 - 3x^2 + 2x, & x \leq -2 \\ x + 2, & -2 < x < 2 \\ x^2 + \sin(x), & x \geq 2 \end{cases}$ continuous on the interval $[-1, 1]$? Why or why not?

2 Computing Limits

2.1 Algebraically

Compute the following limits algebraically, and explain your reasoning when/if you compute a limit by evaluation:

1. $\lim_{x \to 1} \dfrac{x^2 - 1}{x - 1}$

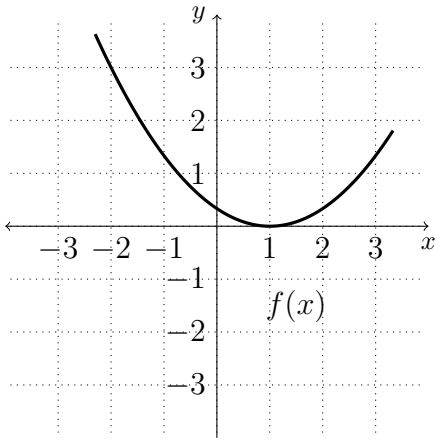
2. $\lim_{x \to 0} \dfrac{(2 + x)^3 - 8}{x}$

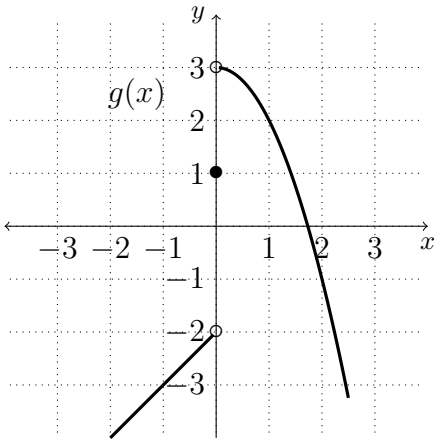
3. $\lim_{x \to 0} \dfrac{\sqrt{x + 1} - 1}{x}$

4. $\lim_{x \to 1} \sin\left(\dfrac{x^2 - 2x + 1}{x - 1}\right)$

2.2 Graphically

Two functions, $f(x)$ and $g(x)$, are graphed below:





(If necessary, you may assume that $f(0) = \frac{1}{3}$.)

Use these graphs (and limit laws) to evaluate each of the following limits. If a limit does not exist, then write "DNE" and explain why the limit does not exist.

(a) $\lim_{x \to 1} \dfrac{1}{2 + f(x)}$

(b) $\lim_{x \to 0} [f(x) + g(x)]$

(c) $\lim_{x \to -1} f(x^2)$

(d) $\lim_{x \to 1} g(f(x))$ (Hint: Be careful!)